

7. A student investigated the reaction between hydrochloric acid (HCl) and calcium carbonate (CaCO_3). She measured the volume of gas produced for different concentrations of acid after 60 seconds.

She started each reaction at the same temperature, with the same mass of calcium carbonate and the same volume of acid. The acid was in excess each time.

The table below shows the time taken to produce 60 cm^3 of CO_2 for each concentration of HCl.

Concentration of HCl (mol/dm^3)	Time to produce 60 cm^3 of CO_2 (s)	Mean rate of reaction (cm^3/s)
0.5	234	0.26
1.0	118	0.51
2.0	58	

- (a) (i) **Complete** the table to two significant figures. [2]
- (ii) Estimate the time taken to produce 60 cm^3 of carbon dioxide if a concentration of 4.0 mol/dm^3 HCl is used. [1]

time = s



- (b) (i) State and explain, in terms of particles, the effect of increasing acid concentration on the rate of reaction. [3]

.....

.....

.....

.....

.....

- (ii) The student noticed that when the experiments ran for 10 minutes, she obtained the same volume of carbon dioxide in each case.

Explain her observation.

[2]

.....

.....

.....

END OF PAPER



Group

O

2

60

LO

4

3

Ω

C

2

1

|| ||

1 H Hydrogen 1										4 He Helium 2									
</																			

relative atomic mass

atomic number